

Relative generation and the generator problem for finite factors

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Abstract

We prove that every type II_1 factor with separable predual is generated by one operator, equivalently by two self-adjoint operators. Combined with the established direct-integral reduction to this case, this gives an affirmative solution of the generator problem for von Neumann algebras with separable predual.

1 Introduction

For a subset S of a von Neumann algebra M , write $W^*(S)$ for the smallest unital von Neumann subalgebra containing S . Thus adjoints are included and generation is taken in the weak operator topology. The generator problem asks whether every von Neumann algebra with separable predual has the form $W^*(x)$ for one bounded operator x . Equivalently, it asks for two self-adjoint generators: their sum $a + ib$ recovers a and b as its real and imaginary parts.

We prove this assertion for type II_1 factors by first establishing a relative generation theorem. For a finite factor with normalized trace τ , put $\|x\|_2 = \tau(x^*x)^{1/2}$. An inclusion $P \subset M$ is *irreducible* when $P' \cap M = \mathbb{C}1$. Write $\mathcal{U}(M)$ for the unitary group of M .

Theorem 1.1. *Let $P \subset M$ be an irreducible inclusion of type II_1 factors, with M of separable predual. The set*

$$\{u \in \mathcal{U}(M) : W^*(P, u) = M\}$$

is a dense G_δ subset of the unitary group $\mathcal{U}(M)$ for the trace 2-norm topology.

Theorem 1.2. *Every type II_1 factor M with separable predual is generated by two self-adjoint elements. Equivalently, there is an $x \in M$ such that $M = W^*(x)$.*

Throughout the paper, *separable* for a finite von Neumann algebra means that its predual is separable. This ensures separability of $L^2(M, \tau)$; it does not require operator-norm separability of M .

Earlier work. The type I, hyperfinite, and properly infinite cases of the generator problem were established by Percy, Suzuki and Saitô, and Wogen, respectively [11, 20, 23]; see also the historical discussion in [19, Section 3]. Willig's direct-integral theorem reduces the general question to type II_1 factors [22, Corollary 2]. Consequently, Theorem 1.2 also gives single generation of every von Neumann algebra with separable predual. This last consequence uses the established direct-integral reduction, whereas our proof below concerns finite factors.

Much of the progress in the finite case has used additional structure. Behncke established single generation for tensor products of two II_1 factors [1]. Popa treated factors with Cartan subalgebras [13]; Ge and Popa treated factors with property Γ and recovered the tensor-product case [6]; and Ge and Shen obtained arithmetic group examples with a small-support refinement [7]. Shen's generator invariant unified these and other positive classes [18]. Dykema, Sinclair, Smith, and White established its exact amplification formula and sharpened its connection with the number of self-adjoint generators [4].

Sherman showed, in particular, that finite generation of every countably generated II_1 factor would suffice for a positive answer [19, Theorem 3.8(3)]. More recently, Gao, Kunnawalkam Elayavalli, Patchell, and Tan proved single generation when a countable unitary generating family lies in one sequential-commutation orbit in an ultrapower: successive members can be joined there by finite chains with Haar intermediate unitaries and commuting adjacent terms [5, Theorem A]. Popa's work on tight decompositions studies stable single generation, which requires every amplification of a factor to be singly generated [17].

Requiring generators to vary continuously with a family of factors leads to a different problem. Hayes constructed a W^* -bundle—a continuous family of tracial von Neumann algebras—over a compact metrizable base, with separable-preidual II_1 fibers, for which no finite family of continuous sections generates every fiber [9, Theorem 1.1].

Relative generation by one unitary is already available for finite-index inclusions, by the downward basic construction used in [4, Lemma 6.2]: the associated Jones projection e generates the extension over the subfactor, and $1 - 2e$ is unitary. Theorem 1.1 applies to every irreducible inclusion and also establishes genericity of the relative generators.

The universal conclusion of Theorem 1.2 also makes the generator invariants G and G_{sa} vanish for every separable-preidual II_1 factor. It implies that Voiculescu's microstates free entropy dimensions δ and δ_0 can change under replacement of a finite self-adjoint von Neumann generating tuple. Section 6 gives these deductions and their precise scope.

The main estimate and the proof. Our argument combines Popa's free-independence construction in tracial ultraproducts [14, 15] with a perturbation whose effect accumulates along a long word. Given a unitary u and a target unitary z , we coordinate a perturbation of u across its repeated occurrences in a word with coefficients from P . With a small parameter α fixed, the perturbation has 2-norm of order $\alpha/\sqrt{\ell}$, where ℓ is the number of occurrences. The resulting word has correlation with a test vector ξ equal to $(\alpha/2)\tau(\xi^*z)$ up to an error of order $\alpha^2\|\xi\|_2 + 1/\ell$.

The essential estimate is uniform in ℓ . An operator-norm bound on each perturbation separately would grow with the length of the product. We instead dominate all coefficients of the nonlinear terms by a free-group convolution operator with nonnegative coefficients. The length-one case of Haagerup's inequality [8, Lemma 1.4] bounds its norm independently of ℓ . This controls the full exponential remainder, including cancellations between words. Proposition 3.2 states the precise bounds. The witness word need only retain a fixed positive correlation as $\ell \rightarrow \infty$. The free variables and perturbation are first constructed in a tracial ultrapower. For each fixed word length, choosing one coordinate of their representing sequences gives the required unitaries in M and P , with an arbitrarily small additional error.

For a fixed vector $\xi \in L^2(M, \tau)$, the norm of its projection onto $L^2(W^*(P, u), \tau)$ is a bounded lower semicontinuous function of u . Baire's theorem supplies common continuity points for a countable dense family of vectors, and contractivity extends continuity to every vector. If $W^*(P, u)$ were proper at such a point, a nonzero bounded vector orthogonal to it would have projection norm zero. The perturbation estimate gives nearby unitaries for which that projection norm has a

fixed positive lower bound, a contradiction.

Finally, Popa's irreducible hyperfinite embedding theorem [12, 16] lets us choose P generated by two self-adjoint elements. Relative generation then gives four self-adjoint generators for every separable-predual II_1 factor. Applying this uniform bound to a half-trace corner, and using an explicit two-by-two matrix encoding, gives Theorem 1.2. This encoding belongs to the established matrix-generation techniques of Douglas–Percy and Wogen [3, 23]; we use the von Neumann algebra form of the matrix encoding in [4, Proposition 1.1].

Section 2 states the exact independence input and constructs the free Haar unitaries required in the ultrapower. Section 3 proves the uniform perturbation estimate. Section 4 carries out the continuity argument and identifies the full relative-generator locus as G_δ . Section 5 gives the corner and matrix construction proving single generation. Section 6 treats the general von Neumann algebra, generator-invariant, and free-entropy consequences.

Notation. We write $\|x\|$ for the operator norm and use the inner product $\langle a, b \rangle = \tau(b^*a)$ on $L^2(M, \tau)$, linear in the first variable. The same trace notation is used on tracial ultrapowers. All subalgebras are unital unless a corner unit is specified. For a von Neumann subalgebra $N \subset M$, E_N denotes the trace-preserving conditional expectation and e_N the orthogonal projection from $L^2(M, \tau)$ onto $L^2(N, \tau)$.

2 Free Haar unitaries in an ultrapower

Throughout this section, $P \subset M$ is an irreducible inclusion of II_1 factors. Fix a free ultrafilter ω on $\mathbb{N}$. We use the tracial ultrapower

$$\mathbf{M} = M^\omega = \ell^\infty(\mathbb{N}, M) / \mathcal{I}_\omega, \quad \mathcal{I}_\omega = \left\{ (b_n) \in \ell^\infty(\mathbb{N}, M) : \lim_{n \rightarrow \omega} \|b_n\|_2 = 0 \right\},$$

and its von Neumann subalgebra $\mathbf{Q} = P^\omega$. The trace of an element represented by (b_n) is $\lim_{n \rightarrow \omega} \tau(b_n)$; we denote this trace, and its associated 2-norm, by the same symbols as on M . We identify M with its constant copy in $\mathbf{M}$.

A unitary t in a tracial algebra is called *Haar* if $\tau(t^n) = 0$ for every nonzero integer n . Two unital subalgebras are *free* if every alternating product of centered elements from the two subalgebras has trace zero. We say that a unitary is free from a unital algebra D when $C^*(t)$ and D are free.

For the perturbation in Section 3, we will choose successive Haar unitaries in $\mathbf{Q}$, each free from the algebra containing the prescribed data and the earlier choices. We will also need this freeness to survive multiplication on either side by unitaries from that algebra. Popa's theorem supplies independence relative to $\mathbf{Q}' \cap \mathbf{M}$, so we first identify that commutant.

Lemma 2.1. *The inclusion $\mathbf{Q} \subset \mathbf{M}$ is irreducible:*

$$\mathbf{Q}' \cap \mathbf{M} = \mathbb{C}1.$$

Proof. Let $b \in \mathbf{Q}' \cap \mathbf{M}$ have a uniformly bounded representative (b_n) , and put

$$\delta_n = \sup_{w \in \mathcal{U}(P)} \|wb_nw^* - b_n\|_2.$$

Then $\lim_{n \rightarrow \omega} \delta_n = 0$. Otherwise there would be an $\varepsilon > 0$ and a set belonging to ω on which $\delta_n > \varepsilon$. Choosing on this set a unitary $w_n \in P$ with $\|w_nb_nw_n^* - b_n\|_2 > \varepsilon$, and setting $w_n = 1$ elsewhere, would give a unitary in $\mathbf{Q}$ that does not commute with b .

For each n , let K_n be the closed convex hull, in $L^2(M, \tau)$, of the elements wb_nw^* , $w \in \mathcal{U}(P)$. The unique element $c_n \in K_n$ of smallest 2-norm is fixed by conjugation by every unitary of P : each such conjugation preserves K_n and its Hilbert space norm. Moreover, c_n belongs to M and $\|c_n\| \leq \|b_n\|$. Indeed, convex combinations from this orbit have this operator norm bound, and an operator-norm-bounded 2-norm limit lies in the same bounded ball of M . To see the latter fact, left multiplication by such a convergent sequence converges strongly on the dense subspace $M \subset L^2(M)$, since

$$\|(x_r - x_s)a\|_2 \leq \|x_r - x_s\|_2 \|a\| \quad (a \in M),$$

and hence converges strongly on all of $L^2(M)$ by the uniform operator norm bound. The strong limit belongs to M , and its value at 1 is the prescribed 2-norm limit.

Thus $c_n \in P' \cap M = \mathbb{C}1$. The trace is constant on K_n and continuous in 2-norm, so $c_n = \tau(b_n)1$. Every element of K_n is at 2-distance at most δ_n from b_n . Consequently

$$\|b_n - \tau(b_n)1\|_2 \leq \delta_n \xrightarrow{n \rightarrow \omega} 0.$$

The bounded scalar sequence $(\tau(b_n))$ has a scalar ultralimit, which represents b . This proves the assertion. $\square$

We use the following special case of Popa's independence theorem. The atomic-relative-commutant case needed here already appears in [14, Lemma 1.4 and Theorem 2.1]; we state the later formulation from [15, Theorem 0.1(a)].

Theorem 2.2 (Popa's independence theorem). *Let $P \subset M$ be an irreducible inclusion of II_1 factors, and put $\mathbf{Q} = P^\omega \subset M^\omega = \mathbf{M}$. For every linear subspace $X \subset \mathbf{M}$ that is separable in 2-norm and satisfies $\tau(x) = 0$ for all $x \in X$, there is a diffuse abelian von Neumann algebra $A \subset \mathbf{Q}$ such that*

$$\tau(x_0 a_1 x_1 \cdots a_m x_m) = 0 \tag{2.1}$$

whenever $m \geq 1$, $x_0 \in X \cup \{1\}$, $x_1, \dots, x_m \in X$, and $a_1, \dots, a_m \in A$ have trace zero.

This is the constant-inclusion case of [15, Theorem 0.1(a)]. The hypothesis there is that no nonzero corner of the coordinate algebra P intertwines into $P' \cap M$ inside M . It holds here because $P' \cap M = \mathbb{C}1$: a nonzero corner of the diffuse factor P cannot embed normally into a finite matrix amplification of the scalars. The conclusion of that theorem is independence relative to $\mathbf{Q}' \cap \mathbf{M}$. Lemma 2.1 identifies this commutant with the scalars, giving (2.1).

Lemma 2.3. *Let $D \subset \mathbf{M}$ be a norm-separable unital C^* -algebra. There is a Haar unitary $t \in \mathbf{Q}$ free from D . In particular,*

$$\tau(d_0 t^{n_1} d_1 \cdots t^{n_j} d_j) = 0 \tag{2.2}$$

for every $j \geq 1$, every $n_1, \dots, n_j \in \mathbb{Z} \setminus \{0\}$, and all $d_0, \dots, d_j \in D$ such that $\tau(d_i) = 0$ for $1 \leq i \leq j-1$. The two endpoints d_0, d_j are arbitrary.

Proof. The subspace $X = D \cap \ker \tau$ is norm separable, hence 2-norm separable. For example, applying $d \mapsto d - \tau(d)1$ to a countable norm-dense subset of D gives a countable norm-dense subset of X . Apply Theorem 2.2 to obtain a diffuse abelian algebra $A \subset \mathbf{Q}$.

We first choose a Haar unitary in A . Diffuseness gives arbitrarily small nonzero subprojections of every nonzero projection: repeatedly split and retain a piece of at most half the trace. Normality of the trace and maximality under inclusion then give subprojections of every prescribed smaller trace. We may therefore choose recursively partitions of 1 by projections

$$\{p_{m,k} : 0 \leq k < 2^m\} \subset A, \quad \tau(p_{m,k}) = 2^{-m},$$

with $p_{m,k} = p_{m+1,2k} + p_{m+1,2k+1}$. The self-adjoint elements

$$h_m = \sum_{k=0}^{2^m-1} \frac{k}{2^m} p_{m,k}$$

satisfy $\|h_{m+1} - h_m\| \leq 2^{-(m+1)}$ and converge in norm to an element $h \in A$. For each continuous function f on $[0, 1]$,

$$\tau(f(h)) = \lim_{m \rightarrow \infty} 2^{-m} \sum_{k=0}^{2^m-1} f(k2^{-m}) = \int_0^1 f(s) ds.$$

It follows that $t = \exp(2\pi i h)$ is Haar.

We next remove the centering requirement at the two endpoints, so that (2.2) can be applied to words with arbitrary outside coefficients. Expand each endpoint into its scalar and centered parts. A term with a centered last endpoint vanishes by (2.1), since every nonzero power of t belongs to A and is centered. A term with a centered first endpoint and scalar last endpoint vanishes after moving the first endpoint to the end by traciality. It remains to consider the word with both endpoints omitted. When $j = 1$ this is the Haar property. When $j \geq 2$, rotate the final power t^{n_j} next to the first power t^{n_1} . If $n_j + n_1 \neq 0$, the resulting word is covered by (2.1). If $n_j + n_1 = 0$, the two powers cancel and leave a word starting and ending with centered elements of D . This word is again covered by (2.1), except when $j = 2$, in which case it is simply the centered element d_1 . This proves (2.2).

Laurent polynomials in t are norm dense in $C^*(t)$. As t is Haar, the centered Laurent polynomials are precisely the linear combinations of its nonzero powers. Expanding an alternating centered word, and using (2.2) followed by norm approximation, proves that t is free from D . $\square$

Lemma 2.4 (Absorption of base-algebra unitaries). *Let D be a unital C^* -subalgebra of a finite tracial von Neumann algebra, and let t be a Haar unitary free from D . If $a, b \in \mathcal{U}(D)$, then both atb and t^* are Haar unitaries free from D . Both satisfy (2.2) in place of t .*

Proof. For any Haar unitary free from D , formula (2.2) follows immediately by expanding the endpoints into scalar and centered parts and applying freeness to each resulting alternating word. Replacing t by t^* merely negates the exponents, so we concentrate on $v = atb$.

Consider a word

$$d_0 v^{n_1} d_1 \cdots v^{n_j} d_j$$

with nonzero integer exponents and centered internal factors $d_1, \dots, d_{j-1}$. Expand each power of v into individual letters t and t^{-1} , with elements of D between them. Within one power of v all these letters have the same sign. A gap between opposite signs can therefore occur only between two successive powers of v . Such a gap equals

$$bd_i b^* \quad \text{for a transition from } t \text{ to } t^{-1}, \quad a^* d_i a \quad \text{for a transition from } t^{-1} \text{ to } t.$$

It is centered by traciality. Every other gap joins equal signs. Expand each of these latter gaps into its scalar and centered parts. In a resulting term, removing a scalar gap only joins letters of equal sign. Thus each block joined across scalar gaps is a nonzero power of t , and consecutive blocks are separated by centered elements of D . The endpoint factors remain arbitrary elements of D . Formula (2.2) shows that every resulting term has trace zero.

We have proved (2.2) for v . Taking $j = 1$ and $d_0 = d_1 = 1$ proves that v is Haar. The general formula and norm approximation by Laurent polynomials then prove its freeness from D . $\square$

We may apply Lemma 2.3 successively. Specifically, given a norm-separable unital C^* -algebra $D_0 \subset \mathbf{M}$ and an integer $r \geq 0$, choose $t_j \in \mathbf{Q}$ Haar and free from D_j , and set

$$D_{j+1} = C^*(D_j, t_j), \quad 0 \leq j \leq r.$$

Each D_{j+1} remains norm separable: polynomials in a countable norm-dense subset of D_j , the two elements t_j, t_j^* , and rational complex coefficients form a countable dense set. Thus every application of the lemma has the required separability hypothesis, even though the previously chosen unitaries already belong to $\mathbf{Q}$.

3 A perturbation with uniform correlation

We seek perturbations that tend to zero in 2-norm while producing a fixed first-order correlation along words of increasing length. The nonlinear error must remain bounded independently of that length. For this purpose we first record the length-one case of Haagerup's inequality [8, Lemma 1.4], together with its elementary proof.

Lemma 3.1. *Let F be a free group with a specified free basis, let $\mathcal{L}$ be the set of basis elements and their inverses, and let λ be the left regular representation on $\ell^2(F)$. For every finitely supported family $(b_\gamma)_{\gamma \in \mathcal{L}}$,*

$$\left\| \sum_{\gamma \in \mathcal{L}} b_\gamma \lambda(\gamma) \right\| \leq 2 \left(\sum_{\gamma \in \mathcal{L}} |b_\gamma|^2 \right)^{1/2}.$$

Proof. For a letter a , let B_a be the orthogonal projection onto the closed span of reduced words beginning with a . Split

$$\lambda(\gamma) = \lambda(\gamma)(1 - B_{\gamma^{-1}}) + \lambda(\gamma)B_{\gamma^{-1}}.$$

The first summand has range in the range of B_γ . These ranges are mutually orthogonal, so for $\eta \in \ell^2(F)$,

$$\begin{aligned} \left\| \sum_{\gamma} b_\gamma \lambda(\gamma)(1 - B_{\gamma^{-1}}) \eta \right\|^2 &= \sum_{\gamma} |b_\gamma|^2 \|(1 - B_{\gamma^{-1}}) \eta\|^2 \\ &\leq \left(\sum_{\gamma} |b_\gamma|^2 \right) \|\eta\|^2. \end{aligned}$$

For the second summands, the triangle inequality and Cauchy-Schwarz give

$$\begin{aligned} \left\| \sum_{\gamma} b_\gamma \lambda(\gamma) B_{\gamma^{-1}} \eta \right\| &\leq \sum_{\gamma} |b_\gamma| \|B_{\gamma^{-1}} \eta\| \\ &\leq \left(\sum_{\gamma} |b_\gamma|^2 \right)^{1/2} \|\eta\|, \end{aligned}$$

because the projections $B_{\gamma^{-1}}$ are mutually orthogonal. Adding the two estimates proves the lemma. $\square$

Proposition 3.2. *Let $P \subset M$ be an irreducible inclusion of Π_1 factors. Given $u, z \in \mathcal{U}(M)$, $\xi \in M$, and $0 < \alpha < 1/2$, for every integer $\ell \geq 1$ there are $\tilde{u} \in \mathcal{U}(M)$ and $\tilde{t}_0, \dots, \tilde{t}_\ell \in \mathcal{U}(P)$ such that, with*

$$\tilde{W} = \tilde{t}_0 \prod_{k=1}^{\ell} (\tilde{u} \tilde{t}_k),$$

one has

$$\begin{aligned}\|\tilde{u} - u\|_2 &\leq \frac{\alpha}{\sqrt{2\ell}} + \frac{1}{\ell}, \\ \left| \tau(\tilde{\xi}^* \tilde{W}) - \frac{\alpha}{2} \tau(\tilde{\xi}^* z) \right| &\leq \|\tilde{\xi}\|_2 \frac{4\alpha^2}{1-2\alpha} + \frac{1}{\ell}.\end{aligned}\tag{3.1}$$

All products indexed by k are taken in increasing order from left to right. The unitaries may depend on ℓ .

Proof. Fix ℓ and work first in $\mathbf{M} = M^\omega$, with $\mathbf{Q} = P^\omega$. We use τ also for the ultrapower trace.

Free variables and prefixes. Set $D_0 = C^*(1, u, z, \xi)$. By Lemma 2.3, choose successively $t_j \in \mathcal{U}(\mathbf{Q})$ Haar and free from D_j , and put $D_{j+1} = C^*(D_j, t_j)$, for $0 \leq j \leq \ell$. Each of these C^* -algebras is norm separable. Define

$$s_1 = t_0, \quad s_{j+1} = ut_j \quad (1 \leq j \leq \ell), \quad g_k = s_1 \cdots s_k \quad (1 \leq k \leq \ell),$$

and

$$W = g_\ell s_{\ell+1} = t_0 \prod_{k=1}^{\ell} (ut_k).$$

Thus g_k is the prefix immediately preceding the k th occurrence of u in W . Since $D_j = C^*(D_0, s_1, \dots, s_j)$, Lemma 2.4 shows that s_{j+1} is Haar and free from D_j . In particular, $s_1, \dots, s_\ell$ are free Haar unitaries. Every nonidentity reduced word in them has trace zero, so the group they generate can be identified with the free group G on these generators. Since $W = (g_\ell u) t_\ell$ and $g_\ell u \in \mathcal{U}(D_\ell)$, the same lemma shows that W is Haar and free from D_ℓ . Applying it to W^* and $z \in \mathcal{U}(D_\ell)$ also shows that

$$y = zW^*$$

is Haar and free from D_ℓ .

For $p \in G$ set $Y_p = py p^{-1}$. These form a free family of Haar unitaries. Indeed, if $a_1, \dots, a_r$ are nonzero integers and successive labels $p_1, \dots, p_r$ are distinct, then

$$Y_{p_1}^{a_1} \cdots Y_{p_r}^{a_r} = p_1 y^{a_1} (p_1^{-1} p_2) y^{a_2} \cdots (p_{r-1}^{-1} p_r) y^{a_r} p_r^{-1}.$$

Each internal ratio is a nonidentity element of G and hence is centered in D_ℓ . Equation (2.2) gives zero trace. It follows, by applying this to the reduction of $v^* w$, that distinct reduced words in the Y_p are orthonormal in $L^2(\mathbf{M}, \tau)$, including the empty word.

A left perturbation of u at its k th occurrence in W is conjugated by g_k when moved to the front of the word. We therefore use the conjugates $g_j^{-1} y g_j$: conjugating the j th one by g_j returns y . Taking imaginary parts will give a self-adjoint perturbation, while freeness makes their average small in 2-norm. Put $\text{Im}(b) = (b - b^*) / (2i)$ and define

$$h = \frac{\alpha}{\ell} \sum_{j=1}^{\ell} \text{Im}(g_j^{-1} y g_j), \quad u' = e^{ih} u, \quad W' = t_0 \prod_{k=1}^{\ell} (u' t_k).$$

The g_j^{-1} are distinct in G , so the summands defining h are orthogonal centered imaginary parts of free Haar unitaries. Each has squared 2-norm $1/2$. Consequently,

$$h = h^*, \quad \|h\| \leq \alpha, \quad \|h\|_2 = \frac{\alpha}{\sqrt{2\ell}}.$$

Write $H_k = g_k h g_k^{-1}$, $p_{kj} = g_k g_j^{-1}$, and $c = \alpha / (2\ell)$. Cancelling consecutive prefixes gives

$$W' = \left(\prod_{k=1}^{\ell} e^{iH_k} \right) W, \quad iH_k = c \sum_{j=1}^{\ell} (Y_{p_{kj}} - Y_{p_{kj}}^{-1}). \quad (3.2)$$

For the first identity, use $e^{iH_k} = g_k e^{ih} g_k^{-1}$ and $g_k^{-1} g_{k+1} = ut_k$ for $1 \leq k < \ell$; the product order is therefore essential.

Let $S = \{p_{kj} : 1 \leq k, j \leq \ell\}$ and let n_p count the ordered pairs (k, j) with $p_{kj} = p$. For the identity e of G , one has $n_e = \ell$. If $k \neq j$, the word

$$p_{kj} = s_1 \cdots s_k s_j^{-1} \cdots s_1^{-1}$$

is reduced, since its joining letters have different indices. Its initial positive segment and its final negative segment determine k and j . All off-diagonal pairs therefore give distinct nonidentity words, and

$$\sum_{p \in S} n_p^2 = \ell^2 + \ell(\ell - 1) = 2\ell^2 - \ell. \quad (3.3)$$

The first-order correlation. Put $C_1 = \sum_{k=1}^{\ell} iH_k$. Each diagonal term $cY_{p_{kk}}W$ equals cz , because $p_{kk} = e$ and $Y_e W = z$. We show that all other linear terms vanish when paired with ζ . Equation (2.2) for W and D_ℓ gives $\tau(\zeta^* W) = 0$. For $p \neq e$, the element p^{-1} is centered in D_ℓ , so

$$\tau(\zeta^* Y_p W) = \tau(\zeta^* p z W^* p^{-1} W) = 0.$$

For every $p \in G$, one also has

$$\tau(\zeta^* Y_p^{-1} W) = \tau(\zeta^* p W z^* p^{-1} W) = 0.$$

Indeed, split the internal factor $z^* p^{-1} \in D_\ell$ into its centered and scalar parts. The first contribution vanishes by (2.2); the second is a scalar multiple of $\tau(\zeta^* p W^2)$ and vanishes by the same formula. It follows that

$$\tau(\zeta^* (1 + C_1) W) = c n_e \tau(\zeta^* z) = \frac{\alpha}{2} \tau(\zeta^* z).$$

The uniform remainder estimate. It remains to bound the contribution of $\prod_k e^{iH_k} - 1 - C_1$ uniformly in ℓ . Group the ordered product of exponentials by its original degree:

$$C_d = \sum_{\substack{m_1 + \cdots + m_\ell = d \\ m_k \geq 0}} \frac{(iH_1)^{m_1} \cdots (iH_\ell)^{m_\ell}}{m_1! \cdots m_\ell!}.$$

The term C_0 is 1, and the degree-one term is the C_1 above. For fixed ℓ , the expansion is absolutely convergent in operator norm, because

$$\sum_{d \geq 0} \|C_d\| \leq \prod_{k=1}^{\ell} \sum_{m \geq 0} \frac{\|H_k\|^m}{m!} = \exp \left(\sum_{k=1}^{\ell} \|H_k\| \right) \leq e^{\ell\alpha}.$$

The Cauchy product may thus be grouped in this way, and its sum is $\prod_k e^{iH_k}$. We will prove the bound

$$\left\| \prod_{k=1}^{\ell} e^{iH_k} - 1 - C_1 \right\|_2 \leq \frac{4\alpha^2}{1 - 2\alpha}, \quad (3.4)$$

which, unlike the preceding norm-convergence bound, is uniform in ℓ .

Let F_S be the formal free group with generators T_p , $p \in S$. In particular T_e is a generator, not the identity of F_S . Let $\Phi : \mathbb{C}[F_S] \rightarrow \mathbf{M}$ be the homomorphism $\Phi(T_p) = Y_p$. By the orthonormality proved above, for every finite group-ring sum $f = \sum_{w \in F_S} f(w)w$ one has

$$\|\Phi(f)\|_2^2 = \sum_{w \in F_S} |f(w)|^2. \quad (3.5)$$

Define

$$B_k = c \sum_{j=1}^{\ell} (T_{p_{kj}} - T_{p_{kj}}^{-1}), \quad A_k = c \sum_{j=1}^{\ell} (T_{p_{kj}} + T_{p_{kj}}^{-1}),$$

and

$$A = \sum_{k=1}^{\ell} A_k = c \sum_{p \in S} n_p (T_p + T_p^{-1}).$$

For a group-ring element f , write $|f|$ for its coefficientwise absolute value, and compare nonnegative coefficient functions coefficientwise. Group multiplication gives

$$|fg|(w) = \left| \sum_{ab=w} f(a)g(b) \right| \leq \sum_{ab=w} |f(a)||g(b)| = (|f||g|)(w).$$

This comparison already includes all free reductions and all contributions collected at the same resulting word. Thus the formal element

$$C_d = \sum_{\substack{m_1 + \dots + m_{\ell} = d \\ m_k \geq 0}} \frac{B_1^{m_1} \dots B_{\ell}^{m_{\ell}}}{m_1! \dots m_{\ell}!}, \quad \Phi(C_d) = C_d,$$

satisfies

$$|C_d| \leq \sum_{m_1 + \dots + m_{\ell} = d} \frac{A_1^{m_1} \dots A_{\ell}^{m_{\ell}}}{m_1! \dots m_{\ell}!} \leq A^d. \quad (3.6)$$

For the last inequality, each tuple $(m_1, \dots, m_{\ell})$ corresponds to one nondecreasing sequence of row indices in the distributive expansion of $(A_1 + \dots + A_{\ell})^d$. Its factorial weight is at most one, and every other summand has nonnegative coefficients. No commutation of the A_k is used. The index d records the degree before reduction; resulting words may have shorter length, including zero.

Let λ now denote the left regular representation of F_S , and let δ_{id} be the unit vector at its identity. By (3.5) and (3.6),

$$\|C_d\|_2 \leq \|A^d\|_{\ell^2(F_S)} = \|\lambda(A)^d \delta_{\text{id}}\|_{\ell^2(F_S)} \leq \|\lambda(A)\|^d.$$

This uses only the coefficient 2-norm identity (3.5), with no comparison of operator norms between the two representations. Lemma 3.1 and (3.3) give

$$\|\lambda(A)\| \leq 2 \left(2c^2 \sum_{p \in S} n_p^2 \right)^{1/2} = 2\alpha \left(1 - \frac{1}{2\ell} \right)^{1/2} \leq 2\alpha < 1.$$

Consequently $\|C_d\|_2 \leq (2\alpha)^d$. The operator-norm convergence established above implies convergence of the same partial sums in 2-norm. Taking their tails and using the triangle inequality therefore yields

$$\left\| \prod_{k=1}^{\ell} e^{iH_k} - 1 - C_1 \right\|_2 \leq \sum_{d \geq 2} (2\alpha)^d = \frac{4\alpha^2}{1-2\alpha}.$$

In particular, orthogonality between different degrees is not required. This proves (3.4).

Combining the first-order correlation with (3.2), (3.4), and Cauchy–Schwarz gives

$$\left| \tau(\xi^* W') - \frac{\alpha}{2} \tau(\xi^* z) \right| \leq \|\xi\|_2 \frac{4\alpha^2}{1-2\alpha}. \quad (3.7)$$

Lifting for the fixed word length. Each t_j has a representative consisting of unitaries in P . To see this, start with a bounded representative (b_n) of any unitary in P^ω . The polar part of b_n extends to a unitary $w_n \in P$, since its initial and final complementary projections have equal trace in the finite factor P . Then $b_n = w_n |b_n|$ and functional calculus gives

$$\|b_n - w_n\|_2 = \||b_n| - 1\|_2 \leq \|b_n^* b_n - 1\|_2 \xrightarrow{\omega} 0.$$

Here the inequality follows from $|\sqrt{t} - 1| \leq |t - 1|$ for $t \geq 0$.

Choose unitary representatives $t_{j,n} \in P$ for the finitely many t_j . Using the constant representatives of u, z , form $g_{k,n}, W_n, y_n, h_n, u'_n, W'_n$ by exactly the formulas above. For this fixed ℓ these are bounded sequences, with $h_n = h_n^*$ and $\|h_n\| \leq \alpha$. Bounded algebraic operations and continuous functional calculus in the quotient show that they represent the corresponding ultrapower elements. In particular,

$$\lim_{n \rightarrow \omega} \|h_n\|_2 = \frac{\alpha}{\sqrt{2\ell}}, \quad \lim_{n \rightarrow \omega} \tau(\xi^* W'_n) = \tau(\xi^* W').$$

Moreover $\|u'_n - u\|_2 \leq \|h_n\|_2$, by the scalar inequality $|e^{it} - 1| \leq |t|$ and functional calculus. Choose one index n for which

$$\|h_n\|_2 < \frac{\alpha}{\sqrt{2\ell}} + \frac{1}{\ell}, \quad |\tau(\xi^* W'_n) - \tau(\xi^* W')| < \frac{1}{\ell}.$$

Both conditions hold on sets belonging to ω , so their intersection is nonempty. Set $\tilde{u} = u'_n$ and $\tilde{t}_j = t_{j,n}$. Equation (3.7) proves (3.1). This choice is made separately for each ℓ , as the proposition permits. $\square$

4 Relative generation by continuity

We now convert the correlation supplied by Proposition 3.2 into generation. Throughout this section, $P \subset M$ is an irreducible inclusion of II_1 factors and M has separable predual.

Lemma 4.1. *The Hilbert space $L^2(M, \tau)$ is separable, and $\mathcal{U}(M)$ is complete for the metric $d(u, v) = \|u - v\|_2$.*

Proof. The unit ball of M is compact and metrizable in the weak-star topology, so it has a countable weak-star dense subset S . For each $\xi \in L^2(M, \tau)$, the functional $a \mapsto \langle a, \xi \rangle$ on M is normal. Indeed,

normal functionals form a norm-closed subspace of M^* , and approximation of ξ by elements of M in L^2 gives norm approximation of this functional by normal functionals, since

$$\sup_{\|a\| \leq 1} |\langle a, \xi - \eta \rangle| \leq \|\xi - \eta\|_2.$$

If ξ is orthogonal to S , normality implies that it is orthogonal to the unit ball, and hence to M . Thus S has dense linear span in $L^2(M, \tau)$, proving separability.

Let (u_n) be a Cauchy sequence in $\mathcal{U}(M)$ for d . The adjoints are also Cauchy, because $\|u_n^* - u_m^*\|_2 = \|u_n - u_m\|_2$. For $a \in M$,

$$\|(u_n - u_m)a\|_2 \leq \|a\| \|u_n - u_m\|_2.$$

The left multiplication operators associated with u_n therefore converge strongly on the dense subspace $M \subset L^2(M, \tau)$. Their uniform operator norm bound extends this convergence to the whole Hilbert space. The same holds for the left multipliers associated with u_n^* . Strong closedness of the left regular representation of M gives limits $u, v \in M$, and passage to inner products gives $v = u^*$. Multiplication is jointly strongly continuous on bounded sets, so $uv = vu = 1$. Finally, convergence on the vector 1 gives $\|u_n - u\|_2 \rightarrow 0$. $\square$

For $u \in \mathcal{U}(M)$, write

$$N_u = W^*(P, u), \quad e_u : L^2(M, \tau) \longrightarrow L^2(N_u, \tau)$$

for the orthogonal projection. For $\xi \in L^2(M, \tau)$, put $f_\xi(u) = \|e_u \xi\|_2$.

If N_u is proper, there is a nonzero bounded vector orthogonal to $L^2(N_u, \tau)$. The perturbation proposition will keep its projection norm bounded away from zero along a sequence of unitaries converging to u , so we seek points at which every f_ξ is continuous simultaneously.

Lemma 4.2. *Each f_ξ is bounded and lower semicontinuous on $(\mathcal{U}(M), d)$. There is a dense G_δ subset $\mathcal{C} \subset \mathcal{U}(M)$ such that every f_ξ , $\xi \in L^2(M, \tau)$, is continuous at every point of $\mathcal{C}$.*

Proof. Let $\mathcal{P}_P$ denote the family of finite $*$ -polynomials in a unitary variable with coefficients in P . For each fixed u , the values $b(u)$, $b \in \mathcal{P}_P$, are dense in $L^2(N_u, \tau)$. To see this, take the operator norm closure of the polynomial algebra. Its von Neumann closure is N_u , and Kaplansky density approximates each element of N_u strongly by bounded elements of this C^* -algebra. Evaluation on $1 \in L^2(M, \tau)$ gives 2-norm approximation, which can then be followed by operator norm approximation by polynomials. The density of N_u in $L^2(N_u, \tau)$ finishes the assertion.

For fixed $b \in \mathcal{P}_P$, the map $u \mapsto b(u)$ is continuous in 2-norm. This follows by telescoping products, using the fixed operator norm bounds of the coefficients, the fact that unitaries have norm one, and the identity $\|u^* - v^*\|_2 = \|u - v\|_2$. Consequently

$$d_\xi(u) := \text{dist}(\xi, L^2(N_u, \tau)) = \inf_{b \in \mathcal{P}_P} \|\xi - b(u)\|_2$$

is upper semicontinuous: an infimum of any family of continuous real functions is upper semicontinuous. Orthogonality gives

$$f_\xi(u)^2 = \|\xi\|_2^2 - d_\xi(u)^2,$$

so f_ξ is lower semicontinuous and $0 \leq f_\xi(u) \leq \|\xi\|_2$.

For completeness, a bounded lower semicontinuous real function f on a complete metric space is continuous on a dense G_δ set. For $\varepsilon > 0$, let G_ε be the union of all open sets on which the oscillation of f is less than ε . This union is dense. Indeed, in any nonempty open set O , write $s = \sup_O f$ and choose $x \in O$ with $f(x) > s - \varepsilon/3$. Lower semicontinuity gives an open neighborhood $V \subset O$ of x on which $f > f(x) - \varepsilon/3$. The oscillation on V is less than ε . Baire's theorem now shows that $\bigcap_{n \geq 1} G_{1/n}$ is dense, and f is continuous at each of its points.

Apply this construction to f_η for η in a countable dense subset of $L^2(M, \tau)$ and intersect the resulting dense G_δ sets. By Baire's theorem, their intersection $\mathcal{C}$ is again a dense G_δ set. For arbitrary $\xi, \eta \in L^2(M, \tau)$, contractivity of e_u gives

$$|f_\xi(u) - f_\eta(u)| \leq \|\xi - \eta\|_2 \quad (u \in \mathcal{U}(M)). \quad (4.1)$$

Thus, for $u \in \mathcal{C}$ and $v \in \mathcal{U}(M)$,

$$|f_\xi(v) - f_\xi(u)| \leq 2\|\xi - \eta\|_2 + |f_\eta(v) - f_\eta(u)|.$$

Approximating ξ by an η from the countable dense subset, and then letting $v \rightarrow u$, proves continuity of every f_ξ at u . $\square$

Proof of Theorem 1.1. Take $u \in \mathcal{C}$ from Lemma 4.2, and suppose that $N_u \neq M$. There is a unitary $z \in M \setminus N_u$, since the unitaries linearly span M . Let $E_{N_u} : M \rightarrow N_u$ be the trace-preserving conditional expectation, and put

$$\xi = z - E_{N_u}(z) \in M, \quad s = \|\xi\|_2 > 0.$$

The conditional expectation is the restriction to M of e_u . Therefore

$$e_u \xi = 0, \quad f_\xi(u) = 0, \quad \tau(\xi^* z) = s^2.$$

Fix

$$0 < \alpha < \min\{1/4, s/32\}. \quad (4.2)$$

For every integer $\ell \geq 1$, Proposition 3.2 provides a unitary $u_\ell \in M$ and unitaries $t_{0,\ell}, \dots, t_{\ell,\ell} \in P$ such that, with

$$W_\ell = t_{0,\ell} \prod_{k=1}^{\ell} (u_\ell t_{k,\ell}) \in \mathcal{U}(N_{u_\ell}),$$

one has

$$\|u_\ell - u\|_2 \leq \frac{\alpha}{\sqrt{2\ell}} + \frac{1}{\ell}, \quad (4.3)$$

$$\left| \tau(\xi^* W_\ell) - \frac{\alpha s^2}{2} \right| \leq s \frac{4\alpha^2}{1-2\alpha} + \frac{1}{\ell}. \quad (4.4)$$

The fixed choice (4.2) gives

$$s \frac{4\alpha^2}{1-2\alpha} < 8s\alpha^2 < \frac{\alpha s^2}{4}.$$

It follows from (4.4) that

$$|\tau(\xi^* W_\ell)| \geq \frac{\alpha s^2}{4} - \frac{1}{\ell} \geq \frac{\alpha s^2}{8}$$

for all sufficiently large ℓ . On the other hand, $\|W_\ell\|_2 = 1$ and $W_\ell \in L^2(N_{u_\ell}, \tau)$, so

$$|\tau(\xi^* W_\ell)| = |\langle W_\ell, e_{u_\ell} \xi \rangle| \leq f_\xi(u_\ell).$$

By (4.3), $u_\ell \rightarrow u$ in 2-norm. Continuity of f_ξ at u gives $f_\xi(u_\ell) \rightarrow f_\xi(u) = 0$, a contradiction. Thus $M = W^*(P, u)$ for every $u \in \mathcal{C}$.

To see that the full set of relative generators is G_δ , choose $(\eta_j)_{j \geq 1}$ dense in $L^2(M, \tau)$. With d_{η_j} as in Lemma 4.2, this set is

$$\bigcap_{j, n \geq 1} \{v \in \mathcal{U}(M) : d_{\eta_j}(v) < 1/n\}.$$

Each set in the intersection is open by upper semicontinuity. Vanishing of every distance is equivalent to $L^2(N_v, \tau) = L^2(M, \tau)$, hence to $N_v = M$ by the trace-preserving conditional expectation. The set is dense because it contains $\mathcal{C}$. $\square$

5 Two self-adjoint generators

Relative generation over an irreducible hyperfinite subfactor will first give four self-adjoint generators for every separable II_1 factor. We then apply this uniform bound to a half-trace corner N of M : the identification $M \cong M_2(N)$ allows the four generators of N to be encoded by two self-adjoint matrices.

We use Popa's irreducible hyperfinite embedding theorem in the following form.

Theorem 5.1 (Popa). *Let $N \subset M$ be an irreducible inclusion of II_1 factors with separable preduals. There is a hyperfinite II_1 subfactor $R \subset N$ such that $R' \cap M = \mathbb{C}1$.*

The special case $N = M$ is [12, Corollary 4.1]; it is also used explicitly in [19, proof of Theorem 3.8]. The relative form above is recalled in [16, Introduction].

The two-generator result for the hyperfinite factor is classical [20]. We give a tensor-model proof for the later count of generators.

Lemma 5.2. *The hyperfinite II_1 factor with separable predual is generated by two self-adjoint elements.*

Proof. By uniqueness of the hyperfinite II_1 factor with separable predual [15, Section 1.3], we may use its standard tensor model

$$R = \overline{\bigotimes_{m \geq 1} (M_2(\mathbb{C}), \text{tr}_2)}.$$

On the m th tensor site, let

$$d_m = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad q_m = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

with the identity on the other sites. On that site, $2d_m q_m (1 - d_m)$ is the matrix unit e_{12} , so d_m, q_m generate the full matrix algebra. Each of the two families (d_m) and (q_m) consists of commuting projections.

For any countable family of commuting projections (e_m) , the operator norm convergent series

$$b = \sum_{m \geq 1} 23^{-m} e_m$$

recovers every e_m by spectral calculus. Inductively, after recovering $e_1, \dots, e_{m-1}$, form

$$r_m = b - \sum_{j < m} 23^{-j} e_j.$$

The restriction of r_m to $1 - e_m$ lies between 0 and 3^{-m} , whereas its restriction to e_m lies between 23^{-m} and 33^{-m} . Since e_m commutes with r_m ,

$$e_m = 1_{((3/2)3^{-m}, \infty)}(r_m) \in W^*(b).$$

Applying this encoding separately to (d_m) and (q_m) gives two self-adjoint elements whose generated von Neumann algebra contains every tensor site, and hence is all of R . $\square$

Proposition 5.3. *Every II_1 factor with separable predual is generated by four self-adjoint elements.*

Proof. Let M be such a factor. By Theorem 5.1, applied with $N = M$, it contains a hyperfinite II_1 subfactor P with $P' \cap M = \mathbb{C}1$. The trace-preserving conditional expectation onto P embeds P_* isometrically into M_* by precomposition, so P has separable predual. Lemma 5.2 gives self-adjoint $a, b \in P$ with $P = W^*(a, b)$, and Theorem 1.1 gives $u \in \mathcal{U}(M)$ with $M = W^*(P, u)$. Hence

$$M = W^*(a, b, \text{Re } u, \text{Im } u). \quad \square$$

The uniform bound in Proposition 5.3 also applies to half-trace corners. The following matrix construction reduces their four generators to two after amplification. It is the von Neumann algebra form of the 2×2 encoding in [4, Proposition 1.1], with the off-diagonal entry fixed to the identity. We include the proof to make the recovery of the generators and matrix units explicit.

Lemma 5.4. *If a unital von Neumann algebra N is generated by four self-adjoint elements, then $M_2(N)$ is generated by two self-adjoint elements.*

Proof. Write $N = W^*(a_1, a_2, a_3, a_4)$ with $a_j = a_j^*$, and choose $C > \|a_1\| + \|a_2\|$. Put

$$A = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 + C1_N \end{pmatrix}, \quad B = \begin{pmatrix} a_3 & 1_N \\ 1_N & a_4 \end{pmatrix}, \quad Q = W^*(A, B).$$

Let E_{ij} be the standard matrix units of $M_2(N)$. The first diagonal block of A has spectrum in $[-\|a_1\|, \|a_1\|]$, and the second has spectrum in $[C - \|a_2\|, C + \|a_2\|]$. For $\|a_1\| < \lambda < C - \|a_2\|$, spectral calculus therefore gives

$$E_{11} = 1_{(-\infty, \lambda)}(A) \in Q, \quad E_{22} = 1 - E_{11} \in Q.$$

The off-diagonal cuts of B give $E_{12} = E_{11}BE_{22} \in Q$ and $E_{21} = E_{12}^* \in Q$. The first corner then contains

$$\begin{aligned} a_1 E_{11} &= E_{11} A E_{11}, & a_2 E_{11} &= E_{12} A E_{21} - C E_{11}, \\ a_3 E_{11} &= E_{11} B E_{11}, & a_4 E_{11} &= E_{12} B E_{21}. \end{aligned}$$

Thus $NE_{11} \subset Q$. Multiplying this corner by the matrix units yields $NE_{ij} \subset Q$ for every i, j , and hence $Q = M_2(N)$. $\square$

Proof of Theorem 1.2. Let M be a II_1 factor with separable predual and normalized trace τ . Choose a projection $p \in M$ with $\tau(p) = 1/2$. Set $N = pMp$, with normalized trace $\tau_N(a) = 2\tau(a)$. This corner is a II_1 factor with separable predual: precomposition with the normal compression $a \mapsto pap$ embeds N_* isometrically into M_* . Since p and $1 - p$ have the same trace, there is a partial isometry $v \in M$ with $v^*v = p$ and $vv^* = 1 - p$. The corresponding matrix units give a normal unital $*$ -isomorphism

$$\Phi : M_2(N) \longrightarrow M, \quad \Phi \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = b_{11} + b_{12}v^* + vb_{21} + vb_{22}v^*.$$

Proposition 5.3, applied to N , and Lemma 5.4 provide self-adjoint $A, B \in M_2(N)$ with $W^*(A, B) = M_2(N)$. Define $x = \Phi(A + iB) \in M$. Since the real and imaginary parts of x are $\Phi(A)$ and $\Phi(B)$, respectively,

$$W^*(x) = W^*(\Phi(A), \Phi(B)) = \Phi(M_2(N)) = M. \quad \square$$

6 Consequences for generation and free entropy

The universal statement of Theorem 1.2 has consequences beyond the relative generation theorem. We first apply classical reductions, then compare two generating tuples of a free group factor.

6.1 General von Neumann algebras and generator invariants

Willig's direct-integral reduction [22, Corollary 2], together with the established type I and properly infinite cases, reduces the generator problem for von Neumann algebras with separable predual to the II_1 factor case. Theorem 1.2 therefore gives single generation for every von Neumann algebra with separable predual.

Let G and G_{sa} be the generator invariants defined in [4, Definition 2.7]. They are nonnegative infima over finite generating sets and finite projection partitions; G_{sa} restricts the generating sets to self-adjoint elements. For every complex II_1 factor M with separable predual and normalized trace, we have

$$G(M) = G_{\text{sa}}(M) = 0.$$

Indeed, by [4, Corollary 5.1], a positive G -value for one such factor would force a nonsingly generated factor in the same class, contrary to the universal assertion of Theorem 1.2. The cited generation convention need not include the unit. To pass to that convention, replace a generator x by $y = x + \lambda 1$ with $\lambda > \|x\|$. The invertible y has support projection 1 in its possibly nonunital von Neumann closure; that closure then contains x and equals M . Nonnegativity gives $G(M) = 0$, and [4, Theorem 5.5] gives $G_{\text{sa}}(M) = 2G(M) = 0$. These values are the defining infima; no generating set or projection partition attaining zero is asserted.

6.2 Dependence of free entropy dimensions on generators

For a bounded finite self-adjoint tuple Z in a tracial von Neumann algebra, let $\delta(Z)$ and $\delta_0(Z)$ denote Voiculescu's microstates free entropy dimension and its modified version, respectively [21, Section 2.4]. The W^* -invariance question asks whether either number is the same for any two finite self-adjoint tuples generating the same tracial von Neumann algebra; see [21, Section 2.6].

Corollary 6.1. *For every integer $n \geq 3$, the factor $L(\mathbb{F}_n)$ with its canonical trace has two finite self-adjoint generating tuples on which both δ and δ_0 take different values. In particular, neither dimension is determined by the generated tracial von Neumann algebra.*

Proof. For a bounded self-adjoint d -tuple Z , one has $\delta(Z) \leq d$ and $\delta_0(Z) \leq d$. Both dimensions equal n for a freely independent variance-one semicircular n -tuple; see [21, Section 2.4(a), (c), and (d)] for δ and [2, Section 5] for δ_0 . The standard semicircular n -tuple generates $L(\mathbb{F}_n)$ as a von Neumann algebra. A self-adjoint pair supplied by Theorem 1.2 generates the same factor and has each dimension at most $2 < n$. These two tuples give the assertion. $\square$

A separate isomorphism argument in *An isomorphism of the free group factors* [10, Corollary 7.1] gives a stronger fixed-factor conclusion: each of δ, δ_0 and the nonmicrostates dimensions δ^*, δ^* attains every integer value $n \geq 2$ on finite self-adjoint W^* -generating tuples of $L(\mathbb{F}_2)$. That conclusion uses the trace-preserving isomorphisms between free group factors established in the cited paper. Corollary 6.1 uses only the single-generation theorem proved here and the standard dimension bounds above.

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